Session-XI: Soil Moisture Retention: January 1st week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	15 min.	Recap.	Small group discussion	Tenth session contents, charts and markers.	Participants recollect the contents discussed in tenth session.
2	45 min.	Observation of LTE and data collection.	Small group discussion	FWS group kit, charts and markers.	Participants observe the changes in experimental and control plots and analyze the data.
3	40 min.	Soil moisture retention methods.	Small group discussion	Charts and markers.	Participants list the different soil moisture retention methods.
4	40 min.	Composting methods.	Large group discussion	Charts and markers.	Participants get knowledge on different methods of composting.
5	60 min.	Short Study- Role of organic manures in water holding capacity.	Demonstration	Trays, water, measuring jar, balance, muslin cloth, rubber band.	Participants discuss the importance of organic manure addition in improving the water holding capacity of soil.
6	15 min.	Group dynamics – Fund mobilization.	Large group discussion	Charts and markers.	Participants understand the need to mobilize fund for sustainability of HUNs.
7	10 min.	Homework- Data collection on functional bore wells, pumping hours, days and discharge.	Large group discussion	Charts and markers.	Participants discuss the need to collect the basic information to study the impact of FWS.
8	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session participants will be able to:

- Recap the previous session,
- Observe LTE and collection of data,
- Discuss various soil moisture retention methods,
- Discuss different methods of composting and the uses,
- Discuss importance of fund mobilization, and
- Discuss the ways of collecting data on functional bore wells, pumping hours, days and average discharge.

Process

1. Recap Time: 15 min.

Objective : Participants recap the contents discussed in the previous session.

Content : Session ten contents

- Observation of LTE and analysis.
- Estimation of water requirement by conventional vs. alternate irrigation practices.
- Importance of linkages with line departments, and
- Short study on usage of organic manures to improve water-holding capacity of soil.

Method : Small group discussion

Materials : Tenth session contents, charts and markers.

Process

- Before the start of the session, display visuals used in the tenth session at appropriate place to aid farmers in recap.
- Welcome participants and start the session with FWS pledge.
- In the large group, encourage each participant to share a content that was discussed in previous session and the learning from that one.
- As you facilitate the recap, ask one of the participants to list the contents on the chart paper.
- Discuss the contents and the learning from the previous session.

2. Observation of LTE and data collection Time: 45 min.

Objective : Participants observe and analyze the Long Term Experiment (LTE).

Content :

- Observation of LTE, and
- Data analysis.

Method : Small group discussion

Materials : FWS group kit, charts and markers.

Process

- Request participants to visit the filed and note their observations.
- Encourage the participants note their observations during the filed visit.

Observation sheet:

Village: Name of experiment:

Farmer: Crop:

Area:

Date	Time required per irrigation	Time taken for filling 100 lit drum (Sec, min and 1 hour)	Volume of water	Draft	Yield	Observation

- Ask the groups to come back to the training spot and analyze the observations made during the field visit.
- Request each group to use HESA (based on the experiment) frame work to present their group work.
- Request collaborator to implement the decisions taken by the large group in the experimental plot.

3. Soil moisture retention methods Time: 40min.

Objective : Participants discuss various soil moisture retention methods.

Content : Soil moisture retention method.

Method : Small group discussion

Materials : Charts, and markers.

Process

- Ask participants about the types of soils available in the HU.
- Write participants responses on a chart paper.
- Pose the following questions in large group.
  - Are there any methods to improve the soil moisture retention capacity?
  - If yes, what are the methods?
- List participant's responses on a chart.

Tips to facilitator:

Soil moisture retention methods:

- Application of farm yard manure (FYM)/ compost,
- Mulching,
- Application of green manures, and
- Application of Vermicompost.

- Request each group to take one soil moisture retention method and discuss about the application process.
- Ask small groups to report their discussions in the large group.
- Summarize the discussions.

Tips to facilitator

Here below are some of the merits and demerits of each method.

I. FYM/compost

Merits:

1. Low cost material.
2. Maintains soil quality under continuous cultivation.
3. Improves the soil moisture retention capacity.

Demerits:

1. Bulk in nature.
2. Less available due to decreased livestock population.

II.Mulching(Paddy straw/husk, groundnut husk, saw dust, dried leaves)

Merits:

1. Wide range of raw materials for mulching.
2. Suppress the weed growth.
3. Improve the soil moisture retention capacity.
4. Some mulching material add nutrients to soil.

Demerits:

1. Some times plant debris (stubs, plant parts) may carry pests and diseases to the existing crop.

III.Green manures (Sun hemp, dhaincha, Leguminous crops)

Merits:

1. Provide large quantities of nitrogen for the soil.
2. Improves topsoil depth, water-holding capacity, nutrient content, and texture of soil.
3. The cover they provide for the soil protects the soil from wind or water erosion.
4. Green manure crops provide generous amounts of high protein fodder for animals.
5. Some green manure crops provide human food (edible beans, peas, and pods).

Demerits:

1. It is suitable for Kharif season, and where water is available in abundance.
2. It should be properly decomposed. Wait for 7-10 days to decompose.

4. Composting methods Time: 40min.

Objective : Participants discuss different methods of composting and its advantages.

Content : Composting methods, and uses.

Method : Large group discussion

Materials : Charts and markers.

Process

- Ask participants the following question.
    - What kind of treatment our fathers or elders applied to increase the fertility of soils or to provide nutritive values to soils?
- Write participants responses on a chart.
- Ask participants to list different composting methods.
- List the participant's responses on the chart.
- Ask participants to share the process of different methods of compost making and explain each method.



Tips to facilitator

Following are some of the methods of preparation

1. Pile method.
2. Pit method.
3. Nadep method.

1. Pile method:

This method of composting can be done on any hard surface of the ground

Materials : Dung (cow, pig or poultry),Green plant waste materials, kitchen waste. This is wet organic matter, Dried organic matter. Take equal proportions of all the three materials.

Process

- Clear the space on the ground.
- Collect all the materials and pile them separately.
- Spread dried leaves to 6" height, sprinkle water (If the dung is dry, it can be laid first. If the dung is wet spread the dried leaves as the first layer).
- Spread the dung 1" high over the dried leaves.
- Spread the green matter 6" high on it. Sprinkle water. Repeat the process till all the material is used and the pile is 4 ft high.
- Cover the pile with plastic paper. It should be air tight. Put weights on the cover so that the paper is not displaced.
- After two days the pile becomes warm, which indicates that the decomposition process has started. To check that a stick can be sent inside the pile and it would be hot at the tip of the stick.
- If the process has not started, it indicates that the pile is not covered properly. The air has dried the water. So the pile has to be done again.
- Mix all the ingredients again and spread the layers, sprinkling water and cover it properly air tight.
- To get the compost to be ready it takes 8 weeks.
- The ready compost is dark brown in color and porous and fine textured.

2. Pit method:

Dig a pit of 2 m X 2 m X 1 m dimensions.

Materials: As in the case of pile method.

Process As in the case of pile method. Fill the pit with the materials and cover it soil.

3. Nadep method:

Construct a tank of 2.5 m X 1.5 m X 1 m dimensions with holes on to the walls

Materials: As in the case of pile method.

Process: As in the case of pile method. Fill the tank with the materials and cover it soil. Wet the bed twice a week with sufficient water.



5. Short Study: Role of organic manures in improving water holding capacity (WHC)

Time: 60 min.

Introduction:

The application of organic manures is decreasing in the recent years. Organic manures not only serve as a source of nutrients but also play an important role in improving the soil's physical condition. Organic manure increases the water holding capacity of soil. This study has been conducted to learn about the effect of organic manures on water holding capacity.

Objective:

At the end of the activity participants will be able to:

- Find out the water retention/holding capacity of different soils.
- Discuss the importance of organic manure addition in improving the water holding capacity of soil.

Method:

- Large group discussion to introduce the topic
- Establishment of the study in small groups

Materials: Trays, water, measuring jar, two types of soil, chimneys/pots, balance for weighing, muslin cloth, rubber band/thread

Procedure

- Introduce and discuss the topic in large group.
- Divide the participants into small groups.

Ask the groups to adopt the following process:

1. Collect two different types of soils (clay loam and sand) of one kg each (care should be taken not to collect clay soil, where addition of organic manures will decreases the WHC and improves the drainage).
2. Divide clay loam soil into two parts of 500 grams each.
3. Take 500 grams of clay soil and add organic matter to the soil in the ratio of 5:1 and adjust the quantity to 500 grams.
4. Repeat the same procedure for sandy soil also.
5. Now there are four types of soils i.e., Clay loam, clay loam with organic manure, sandy soil and sandy soil with organic manure.
6. Take four chimneys or transparent bottles with both open.
7. Cover the bottom side of the chimney/bottle by using muslin cloth and tie with rubber band.
8. Now pour the four types of soils into four chimneys/bottles.
9. Keep all the chimneys in the tray.
10. Add 500ml of water to each chimney/bottle and keep the set up for 30 minutes to allow the water to leach down into tray.
11. After 30 minutes collect the water from the trays and measure the quantity by using a measuring jar.
12. Calculate the percentage of water held in the soil / soil + organic matter.

Accordingly each group observed and recorded their observations and shared the results in large group.

6. Group dynamics- Fund mobilization Time: 15 min.

Objective : Participants understand the importance of Fund mobilization.

Content : Importance of Fund mobilization.

Method : Story.

Materials : Script of the story, Charts and markers.

Process

- Long ago, there lived three families in a village.
- Ramana, Surya and Krishna were the Head's of the three families respectively.
- One day, a Bank officer happened to visit their village and offered a certain amount as loan to each family towards the government's initiative to help families progress.
- All the three families were awarded the loan based on their poor financial condition.
- Ramana's family was very happy seeing the amount at a time. They divided the amount equally and spent the amount lavishly in a short time without thinking that loans are meant to be returned.
- Surya and his family members did not understand what to do with the money and kept the entire amount in the cupboard. One day, a thief broke into their house and stole the amount. The family could not repay the loan to the bank.

- Krishna on the other hand held a meeting with his family members and decided to start mushroom cultivation at home along with his regular agriculture activities. All the family members supported him and helped him in the initiative.
- Their business eventually started growing. They paid the loan within the deadline and the bank officer this time visited them and readily agreed to give a larger amount as loan to start a new livelihood.
- Krishna's family expanded their agriculture activities, eventually acquired more land and even bought a new house. They invited Ramana and Surya's families for their house warming ceremony.

After narrating the story, pose the following question in large group:

- "Which family utilized the loan amount in the best possible way?"
- List out the participants responses on a chart paper
- Summarize the discussion by stating that "any institution will run successfully when funds are mobilized and utilized carefully to expand businesses."

7. Home work: Data collection on number of functional bore wells, pumping hours, pumping days and discharge Time: 10 min.

Objective : Participants discuss ways of collecting data on functional bore wells, pumping hours, days, and discharge.

Content : Ways of collecting data on functional bore wells, pumping hours, days, and rate of discharge.

Method : Large group discussion

Materials : Charts and markers.

Process

- Start the discussion with the following words: "In each session we are conducting HESA. In next session we are going to conduct HESA on PHM data. Ask the following questions in large group."

a. What is the data required to conduct HESA?

b. How to collect the data?

Tips to facilitator

- The following data is required to estimate the draft for the Rabi season.
  - Number of functional bore wells,
  - Number of functional days for each bore well,
  - Number of functional hours in each day,
  - Drum discharge (litres per hour).
- It is better to collect number of bore wells connected to each transformer.
- Number of functional days and hours per day of each bore should be collected from the owner of the borewell.
- Drum discharge of an observation borewell in that village.
- GMC services should be utilized to collect the data.

8. Evaluation Time: 10min.

Ask the following questions to evaluate the day's learning. Pose the questions to the large group:

- Can you list the different types of soil moisture retention methods?
- Can you list different kinds of composting methods?
- What are the advantages of compost?